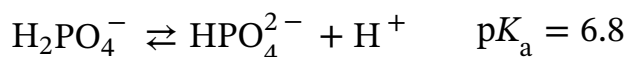


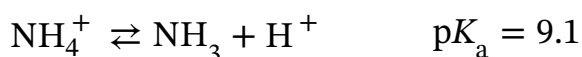
Within the renal system, two buffers operate to counter the production of excess H^+ from luminal H_2CO_3 . They are located within the tubular lumen and maintain urinary pH levels between 4.5 and 8.0.

The first of these is a phosphate buffer, which serves as the primary buffer for urine because its pK_a is very close to the extracellular fluid pH of 7.4. The phosphate buffer operates according to Reaction 1. H_2PO_4^- is excreted in urine.



Reaction 1

During acidosis when tubular pH is low, the phosphate buffer system becomes saturated. An ammonia buffer system then becomes the primary buffer, and NH_4^+ is excreted in urine.



Reaction 2

Titrateable acid (TA) is the total acid in urine, and it can be found using Folin's method. Under normal physiological conditions, TA is approximately equal to the amount of phosphate in urine, which is a product of serum phosphate concentration and glomerular filtration rate. This makes TA a good indicator of kidney function.

Folin's method involves back titration of a urine sample to a pH of 7.4 using a strong base. In a typical procedure, potassium oxalate is added to remove calcium from the sample prior to titration with 1.0 M NaOH. The titration continues until the solution reaches a pH of 7.4 (measured by a pH meter). The amount of base required for the titration can then be used to calculate the amount of H_2PO_4^- in the urine sample.

A phosphate buffer system is made according to Reaction 1 using 5 M H_2PO_4^- and 0.5 M HPO_4^{2-} . Will the pH of this system fall within the pH range for urine?

- ✔ ☒ A. Yes; pH = 5.8 [57%]
☐ B. Yes; pH = 6.8 [28%]
☐ C. No; pH = 8.1 [10%]
☐ D. No; pH = 9.1 [4%]

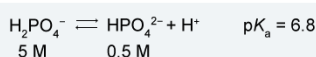
Correct

56% answered correctly

Explanation:

Application of the Henderson-Hasselbalch equation

Given concentration
and pK_a



Use the Henderson-
Hasselbalch equation
to solve for pH.

$$\text{pH} = pK_a + \log \frac{A^-}{HA}$$

5.8 is between
4.5 and 8.0

$$\text{pH} = 6.8 + \log \frac{0.5}{5} = 6.8 + (-1.0) = 5.8$$

A buffer solution is made of weak acid and its conjugate base.

The log of the ratio of base-to-acid determines how much pH differs from pK_a .

©UWorld

A **buffer system** resists changes in **pH** when OH^- or H^+ is added to the solution. Buffer systems accomplish this by containing large amounts of a mixture of either a **weak acid** (HA) with its **conjugate base** (A^-) or a mixture of a **weak base** (B) with its **conjugate acid** (HB^+), which can **neutralize** lesser amounts of any additional H^+ or OH^- that are introduced into the system.

The **Henderson-Hasselbalch equation** is used to find the pH of a buffer system. This equation comes from the equilibrium relationship for the dissociation of a weak acid:



As such, the **acid dissociation constant** K_a is described by the equilibrium equation:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

Rearranging this equation gives:

$$[\text{H}^+] = K_a \frac{[\text{HA}]}{[\text{A}^-]}$$

Because **pH** = $-\log[\text{H}^+]$ and **pK_a** = $-\log K_a$, taking the negative of the logarithm ($-\log$) of both sides of the equation and applying logarithm rules gives the **Henderson-Hasselbalch equation**:

$$\text{pH} = pK_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

For the phosphate buffer system (Reaction 1), the pK_a is 6.8. The **concentration** of H_2PO_4^- (HA) is 5 M and the concentration of HPO_4^{2-} (A^-) is 0.5 M. Substituting

these values into the Henderson-Hasselbalch equation gives:

$$\text{pH} = 6.8 + \log\left(\frac{0.5}{5}\right) = 6.8 + \log(0.1)$$

$$\text{pH} = 6.8 - 1.0 = 5.8$$

Therefore, the pH of the buffer system is 5.8, which is between 4.5 and 8.0 (ie, the acceptable pH range for urine). Note that when evaluating logarithms:

$$\log_a\left(\frac{1}{a}\right) = -\log_a(a) = -1$$

$$\log_{10}\left(\frac{0.5}{5}\right) = \log_{10}\left(\frac{1}{10}\right) = -\log_{10}(10) = -1.0$$

(Choice B) A value of 6.8 is obtained by assuming the pH is the same as the $\text{p}K_a$. In a buffer system, the pH differs from the $\text{p}K_a$ based on the ratio of the concentrations of a weak acid and its conjugate base.

(Choice C) A value of 8.1 is obtained if the $\text{p}K_a$ of the ammonia buffer system is plugged into the Henderson-Hasselbalch equation rather than the $\text{p}K_a$ of the phosphate buffer system.

(Choice D) A value of 9.1 is obtained if the $\text{p}K_a$ is assumed to be the pH of the system and the ammonia buffer system is used instead of the phosphate buffer system.

Educational objective:

A buffer resists changes to pH. A buffer system is comprised of a weak acid (or weak base) and its conjugate. The pH of a particular buffer system can be found using the Henderson-Hasselbalch equation.

Time spent: 0 sec

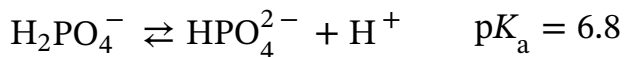
Subject:
General Chemistry

QID: 402312

System:
Solutions and Electrochemistry

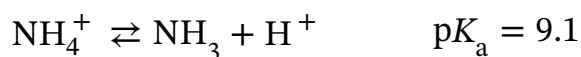
Within the renal system, two buffers operate to counter the production of excess H^+ from luminal H_2CO_3 . They are located within the tubular lumen and maintain urinary pH levels between 4.5 and 8.0.

The first of these is a phosphate buffer, which serves as the primary buffer for urine because its pK_a is very close to the extracellular fluid pH of 7.4. The phosphate buffer operates according to Reaction 1. H_2PO_4^- is excreted in urine.



Reaction 1

During acidosis when tubular pH is low, the phosphate buffer system becomes saturated. An ammonia buffer system then becomes the primary buffer, and NH_4^+ is excreted in urine.



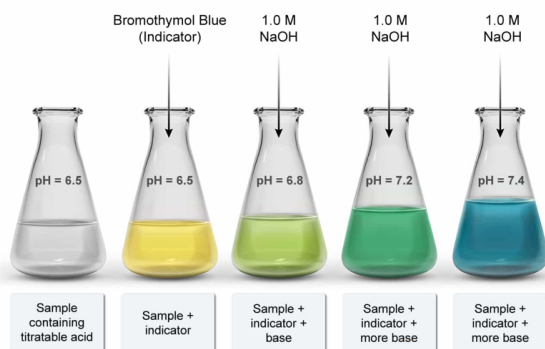
Reaction 2

Titrateable acid (TA) is the total acid in urine, and it can be found using Folin's method. Under normal physiological conditions, TA is approximately equal to the amount of phosphate in urine, which is a product of serum phosphate concentration and glomerular filtration rate. This makes TA a good indicator of kidney function.

Folin's method involves back titration of a urine sample to a pH of 7.4 using a strong base. In a typical procedure, potassium oxalate is added to remove calcium from the sample prior to titration with 1.0 M NaOH. The titration continues until the solution reaches a pH of 7.4 (measured by a pH meter). The amount of base required for the titration can then be used to calculate the amount of H_2PO_4^- in the urine sample.

If a student adds a few drops of bromothymol blue solution ($\text{pK}_a = 7.1$) as an indicator to determine the amount of titrateable acid in 25 mL of urine according to Folin's method, what effect will the bromothymol blue have on the pH of the urine sample?

- ☐ A. It will substantially increase the pH of the sample. [7%]
- ☐ B. It will substantially decrease the pH of the sample. [5%]
- ☐ C. It will neutralize the sample. [6%]
- ☒ D. It will have little effect on the pH of the sample. [80%]

Explanation:

Indicators change color with changing pH but have no effect on the pH of the solution.

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An **indicator** (typically a weak acid or base) is used to visually detect the **endpoint** of an **acid-base titration**. A good indicator must not react with the substance being titrated and should change color close to the **equivalence point** of a titration. The equivalence point is the stoichiometric amount of titrant needed to react with all the solution being titrated. In the case of a buffer solution, a titration is **halfway** to the equivalence point when a pH is achieved that is near the pK_a value of the buffer's acid.

Although indicators are weak acids or bases, their concentration is relatively low (ie, used in small amounts) during a titration. Bromothymol blue is a weakly acidic indicator that changes from pale yellow at pH 6 to dark blue at pH 8 but because only a few drops are added to the titrated solution, it will have little to no effect on the pH of the urine sample.

(Choices A and B) An indicator has little to no effect on the pH of the system because it does not participate in the reaction and is added only in small amounts. The pH neither substantially increases nor decreases.

(Choice C) An indicator only *responds* to changes in pH; it does not aid in neutralizing the solution.

Educational objective:

Indicators are used in acid-base titrations to determine the endpoint of a titration. A good indicator should change color close to the equivalence point, the point at which the stoichiometric amount of titrant needed to consume all the acid or base in the solution has been added. Indicators do not participate in the reaction itself.

Time spent: 0 sec

Subject:

General Chemistry

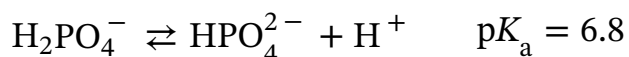
QID: 402313

System:

Solutions and Electrochemistry

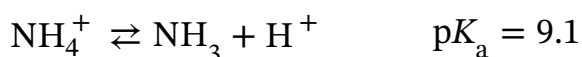
Within the renal system, two buffers operate to counter the production of excess H^+ from luminal H_2CO_3 . They are located within the tubular lumen and maintain urinary pH levels between 4.5 and 8.0.

The first of these is a phosphate buffer, which serves as the primary buffer for urine because its pK_a is very close to the extracellular fluid pH of 7.4. The phosphate buffer operates according to Reaction 1. H_2PO_4^- is excreted in urine.



Reaction 1

During acidosis when tubular pH is low, the phosphate buffer system becomes saturated. An ammonia buffer system then becomes the primary buffer, and NH_4^+ is excreted in urine.



Reaction 2

Titrateable acid (TA) is the total acid in urine, and it can be found using Folin's method. Under normal physiological conditions, TA is approximately equal to the amount of phosphate in urine, which is a product of serum phosphate concentration and glomerular filtration rate. This makes TA a good indicator of kidney function.

Folin's method involves back titration of a urine sample to a pH of 7.4 using a strong base. In a typical procedure, potassium oxalate is added to remove calcium from the sample prior to titration with 1.0 M NaOH. The titration continues until the solution reaches a pH of 7.4 (measured by a pH meter). The amount of base required for the titration can then be used to calculate the amount of H_2PO_4^- in the urine sample.

The chemical process that occurs during acidosis can be replicated *in vitro* by adding a strong acid to an ammonia buffer solution. How will the pH of the solution change if 1.0 mL of 0.5 M H_2SO_4 is added to a 100 mL solution containing equal amounts of 1 M NH_3 and 1 M NH_4^+ ?

- ☐ A. The pH will increase slightly (< 0.1 pH units) because NH_3 and NH_4^+ are a conjugate acid/base pair. [12%]
- ✔ ☒ B. The pH will decrease slightly (< 0.1 pH units) because NH_3 and NH_4^+ act as

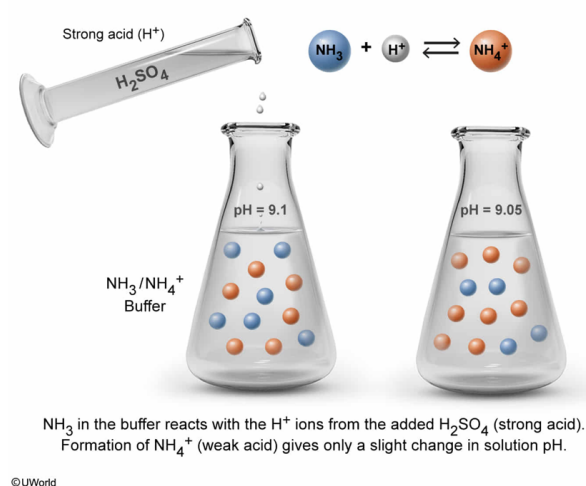
a buffer. [62%]

- ☐ C. The pH will increase significantly (> 1.0 pH units) because NH_3 is a strong base. [3%]
- ☐ D. The pH will decrease significantly (> 1.0 pH units) because H_2SO_4 is a strong acid. [21%]

Correct

63% answered correctly

Explanation:



Buffer systems are comprised of a **weak acid** and its **conjugate base** or a **weak base** and its **conjugate acid** (often in equal concentrations). In either type of buffer, the mixture provides both acidic and basic species to counteract H^+ or OH^- ions from any **strong acid** or **strong base** that is added to the solution. This setup results in a much **smaller change** in **pH** than would otherwise occur without the buffer. The buffer species must be in a much **greater concentration** than the amount of strong acid or strong base that is added to the system. If too much of a strong acid or base is added, the weak acid or conjugate base component of the buffer will be fully consumed, and the buffer will no longer resist changes to pH.

In the experiment described in the question, a strong acid (H_2SO_4) is added to an ammonia buffer system. The amount of H_2SO_4 added is much less than the concentration of NH_3 and will react with the NH_3 to form NH_4^+ (a weak acid); therefore, the pH will decrease slightly but the ammonia buffer system resists a large change in pH.

(Choice A) Protons from H_2SO_4 react with NH_3 (a weak base) to form NH_4^+ (a weak acid). This will cause the pH of the system to *decrease* slightly rather than increase.

(Choice C) NH_3 is a *weak* base rather than a strong base (strong bases tend to form OH^- ions). The addition of an acid will cause the pH to *decrease*, not increase.

(Choice D) Although the pH decreases, it will do so only slightly because a sufficient concentration of NH_3 exists in solution to use the H^+ (added from H_2SO_4) to form NH_4^+ (a weak acid).

Educational objective:

Buffer systems consist of a mixture of a weak acid (or base) and its conjugate that counteract the addition of H^+ and OH^- ions from strong acids or bases, which causes the buffer to resist large changes in pH. Adding a strong acid to a buffer causes the pH to decrease slightly whereas adding a strong base causes the pH to increase slightly.

Time spent:0 sec

Subject:

General Chemistry

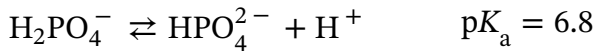
QID:402314

System:

Solutions and Electrochemistry

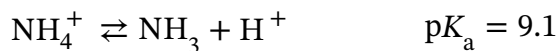
Within the renal system, two buffers operate to counter the production of excess H^+ from luminal H_2CO_3 . They are located within the tubular lumen and maintain urinary pH levels between 4.5 and 8.0.

The first of these is a phosphate buffer, which serves as the primary buffer for urine because its pK_a is very close to the extracellular fluid pH of 7.4. The phosphate buffer operates according to Reaction 1. H_2PO_4^- is excreted in urine.



Reaction 1

During acidosis when tubular pH is low, the phosphate buffer system becomes saturated. An ammonia buffer system then becomes the primary buffer, and NH_4^+ is excreted in urine.



Reaction 2

Titrateable acid (TA) is the total acid in urine, and it can be found using Folin's method. Under normal physiological conditions, TA is approximately equal to the amount of phosphate in urine, which is a product of serum phosphate concentration and glomerular filtration rate. This makes TA a good indicator of kidney function.

Folin's method involves back titration of a urine sample to a pH of 7.4 using a strong base. In a typical procedure, potassium oxalate is added to remove calcium from the sample prior to titration with 1.0 M NaOH. The titration continues until the solution reaches a pH of 7.4 (measured by a pH meter). The amount of base required for the titration can then be used to calculate the amount of H_2PO_4^- in the urine sample.

If the titration of H_2PO_4^- in a urine sample is continued until all the acidic protons in the solution are neutralized, how many equivalents of NaOH are needed to complete the titration of the solution?

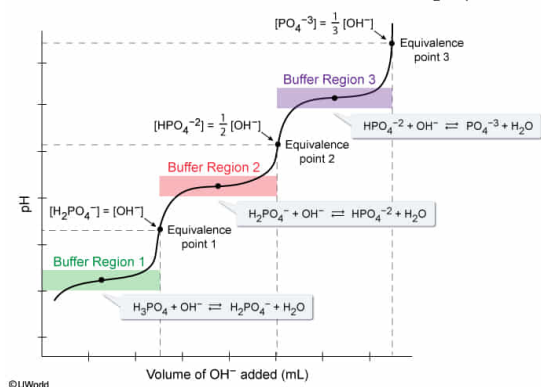
- ☐ A. 1 [20%]
- ✓ ☒ B. 2 [68%]
- ☐ C. 3 [7%]
- ☐ D. 4 [3%]

Correct

69% answered correctly

Explanation:

2 Equivalents of NaOH are needed to neutralize H_2PO_4^-



Polyprotic acids are acids with more than one **acidic proton** (H^+ ion) that can dissociate in water. Each acidic proton dissociates in a stepwise progression that generates a new conjugate base with one less H^+ ion until all acidic protons have dissociated. When a polyprotic acid is **titrated** with a **strong base** (OH^-), **one equivalent** of base is needed to **neutralize each acidic proton**. For example, one equivalent of base neutralizes the first acidic proton in the generic polyprotic acid H_2A :



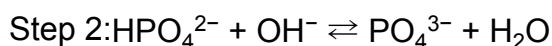
A second equivalent of base neutralizes the acidic proton in HA^- (ie, the conjugate base of H_2A):



During each step of the titration, a transitory **buffer system** is formed as some of the unreacted acid remains with an increasing amount of its conjugate base that is being formed via neutralization. On a **titration curve**, this buffer system is indicated by a plateau where the change in the pH of the solution is minimal (ie, a **buffer region**).

When enough OH^- titrant is added to consume all of a given acidic species, the pH increases rapidly and the titration curve shows a segment with a very steep slope. The midpoint of this nearly vertical segment on the curve is the **equivalence point** (ie, the point where moles of OH^- equal moles of the given acid species). The titration curve of a polyprotic acid will show **one equivalence point for each acidic proton**.

In the passage, H_2PO_4^- (a polyprotic acid) is titrated with NaOH (a strong base). To fully neutralize H_2PO_4^- (ie, no H^+ ions remaining), two acidic protons must be neutralized:



Therefore, a total of two equivalents of NaOH are required to fully neutralize H_2PO_4^- (ie, one OH^- from NaOH for each H^+).

(Choice A) H_2PO_4^- contains two H^+ ions. One equivalent of NaOH will neutralize only one of the H^+ ions.

(Choices C and D) Because H_2PO_4^- contains only two H^+ ions, only two equivalents of NaOH are necessary to fully neutralize the H^+ ions. Additional equivalents are in excess of the stoichiometric amount of base needed.

Educational objective:

Polyprotic acids have more than one proton that can dissociate in water. The titration curve of a polyprotic acid will show as many plateaus and steep climbs as the number of acidic protons in the acid, representing the buffer regions and equivalence points in a titration, respectively.

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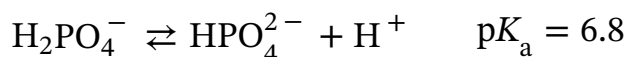
Subject:
General Chemistry

QID:402315

System:
Solutions and Electrochemistry

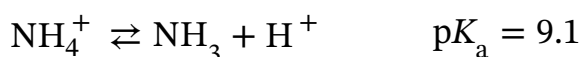
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Reaction 1

During acidosis when tubular pH is low, the phosphate buffer system becomes saturated. An ammonia buffer system then becomes the primary buffer, and NH_4^+ is excreted in urine.



Reaction 2

Titrateable acid (TA) is the total acid in urine, and it can be found using Folin's method. Under normal physiological conditions, TA is approximately equal to the amount of phosphate in urine, which is a product of serum phosphate concentration and glomerular filtration rate. This makes TA a good indicator of kidney function.

Folin's method involves back titration of a urine sample to a pH of 7.4 using a strong base. In a typical procedure, potassium oxalate is added to remove calcium from the sample prior to titration with 1.0 M NaOH. The titration continues until the solution reaches a pH of 7.4 (measured by a pH meter). The amount of base required for the titration can then be used to calculate the amount of H_2PO_4^- in the urine sample.

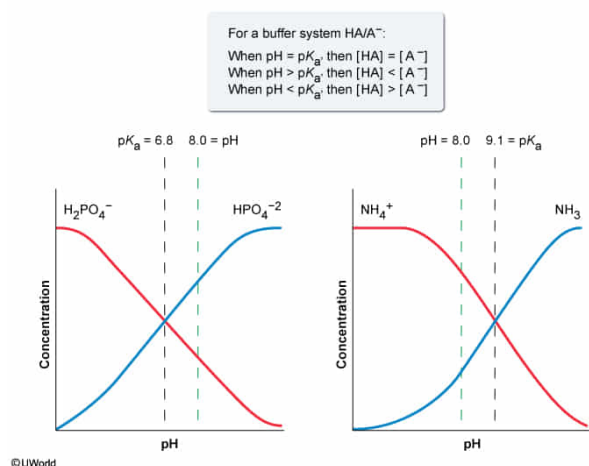
What are the predominant species in the phosphate buffer system and in the ammonia buffer system in an aqueous solution at a pH of 8.0?

- ☐ A. H_2PO_4^- and NH_4^+ [12%]
- ☐ B. H_2PO_4^- and NH_3 [15%]
- ☒ C. HPO_4^{2-} and NH_4^+ [52%]
- ☐ D. HPO_4^{2-} and NH_3 [18%]

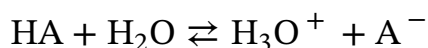
Correct

53% answered correctly

Explanation:



The **dissociation of an acid (HA)** in water can be represented by the general **equilibrium** reaction:



The **relative strength** of an acid (ie, the acidity of a particular hydrogen atom in a molecule) is described by the **acid dissociation constant** K_a , which is also expressed as the $\text{p}K_a$:

$$\text{p}K_a = -\log K_a$$

A solution's acidity is measured by its **pH**, which depends on the **molar concentration** of **hydronium ions** $[\text{H}_3\text{O}^+]$ (often expressed as $[\text{H}^+]$) present in the solution:

$$\text{pH} = -\log[\text{H}^+]$$

The relationship between $\text{p}K_a$ and pH can be quantitatively described by the **derivation** of the Henderson-Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

When the pH of a solution is **equal** to the $\text{p}K_a$ of a weak acid, **equal amounts** of the weak acid and its conjugate base are present in the solution (ie, $[\text{HA}] = [\text{A}^-]$). If the pH is **less than** the $\text{p}K_a$ (ie, the solution is more acidic), the equilibrium is shifted toward a higher concentration of the **protonated** acid (ie, $[\text{HA}] > [\text{A}^-]$). Conversely, if the pH is **greater than** the $\text{p}K_a$ (ie, the solution is more basic), the equilibrium is

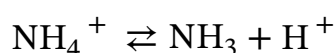
shifted toward a higher concentration of the deprotonated **conjugate base** (ie, $[HA] < [A^-]$).

Accordingly, consider the equilibrium reaction for the phosphate buffer system (Reaction 1):



At a pH of 8.0, the conjugate base HPO_4^{2-} is the dominant species in the phosphate buffer (ie, $[\text{H}_2\text{PO}_4^-] < [\text{HPO}_4^{2-}]$) because the $\text{p}K_a$ of H_2PO_4^- is 6.8, which is less than 8.0.

Likewise, consider the equilibrium reaction for the ammonia buffer system (Reaction 2):



The acidic NH_4^+ is the dominant species in the ammonia buffer (ie, $[\text{NH}_4^+] > [\text{NH}_3]$) because the $\text{p}K_a$ of NH_4^+ is 9.1, which is greater than a pH of 8.0.

(Choices A and B) The $\text{p}K_a$ of H_2PO_4^- is less than the pH of the solution, so H_2PO_4^- is deprotonated to form HPO_4^{2-} as the predominant species.

(Choice D) The $\text{p}K_a$ of NH_4^+ is greater than the pH of the solution, so NH_4^+ remains protonated as the predominant species, *not* NH_3 .

Educational objective:

The $\text{p}K_a$ is a measure of the acidity of a particular proton in a molecule. when the $\text{p}K_a$ is equal to the pH in a buffer system, the amount of unprotonated acid (HA) is equal to the amount of deprotonated conjugate base (A^-), and $[HA] = [A^-]$. If $\text{pH} < \text{p}K_a$, then $[HA] > [A^-]$; if $\text{pH} > \text{p}K_a$, then $[HA] < [A^-]$.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402316

System:
Solutions and Electrochemistry